22.11.2025

Bilkent University Fall 2025

EEE412 Microwave Electronics-Term Project Part 1 Report

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This Project focuses on design the assigned amplifier using a single BFU520W (a BJT). Choosing the operating point according to gain or noise considerations then designing stabilization network(s), input and output matching networks considering the assigned center frequency (0.68Ghz in my case) of the proposed circuit. Below schematic is the overall amplifier design on ADS with the proper input and output matching networks.

DC operating points selected in order to lower the Noise after checking the datasheet,

noise verification problem!

Inital selection didnt work a new calculation and verification needed

changed to,

$VCC = 10V \quad VCE \approx 2V \quad I_C \approx 5mA$ lower the IC to a few milli amps!

$$VCE = \frac{R_1 + R_2}{R_2} \cdot V_0 \rightarrow \approx 1V$$

$$R_C = \frac{VCC - VCE}{I_E + 10 I_B}$$

$$R_C = \frac{10 - 2}{5.1 \times 10^{-3} + 10 \times 57.7 \times 10^{-6}} = 1768\ \Omega$$

picked $1.5K\ \Omega$

when $I_C \approx 5mA \rightarrow$ no problem with stability

$$I_C = \beta \left[VCC - V_0 \left(1 + \frac{(R_C + R_1)}{R_2} \right) \right] \Big/ \left[(\beta+1) R_C + R_1 \right]$$

$$5 \times 10^{-3} = 95 \left[10 - 819 \times 10^{-3} \left(1 + \frac{\overset{1.5K}{R_C} + \overset{8.2K}{R_1}}{\underset{8.2K}{R_2}} \right) \right] \Big/ \left[\overset{1326.5}{96 \times R_C} + \underset{8.2K}{R_1} \right]$$

$\approx 5.1mA$

$$VCE = VCC - \left[I_C (\beta+1)/\beta + \frac{V_0}{R_2} \right] R_C$$

$$\underset{2V}{VCE} = 10 - \left[5 \times 10^{-3} (96)/95 + \frac{\overset{1V}{V_0}}{\underset{8.2K}{R_2}} \right] R_C \rightarrow 1.5K$$

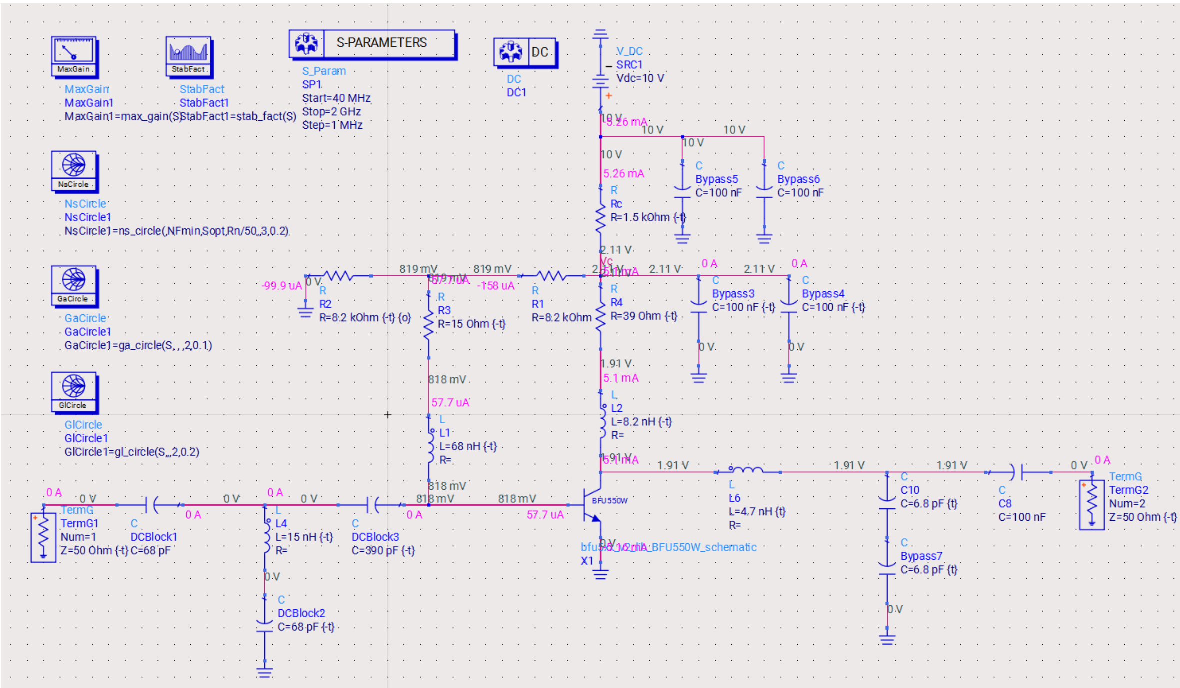

Figure 1: Low noise Amplifier design with the Center Frequency 680Mhz

The amplifier has to be unconditionally stable at all frequencies (40MHz-2GHz), therefore we should check K (stability factor). Figure 2 shows the amplifier is unconditionally stable at all those frequencies.

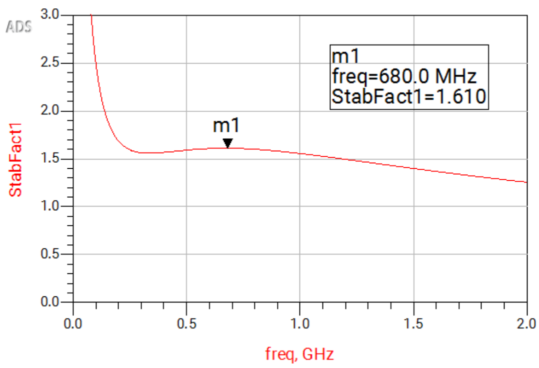

Figure 2: K >1 for all Frequencies

- Center Frequency assigned = **680MHz**
- Bandwidth: 10% of the center frequency
    - **Lower limit: 646 MHz, Upper limit: 714 MHz**
- Gain > 16 – 6.5×f (GHz) dB (f is the center frequency)
    - **Gain > 16-6.5 × 0.68 = 11.58 dB**
- Noise Figure < 0.6 + 0.33×f (GHz) dB (f is the center frequency)
    - **NF < 0.6 + 0.33 × 0.68 = 0.8244 dB**

While I were designing I simultaneously check MAG and Noise Figure, The available gain has to be little higher than our expected gain so that, the real transfered value will be enough, if MAG is already lower than t-our excpected value we cannot reach a higher value whatever we do.

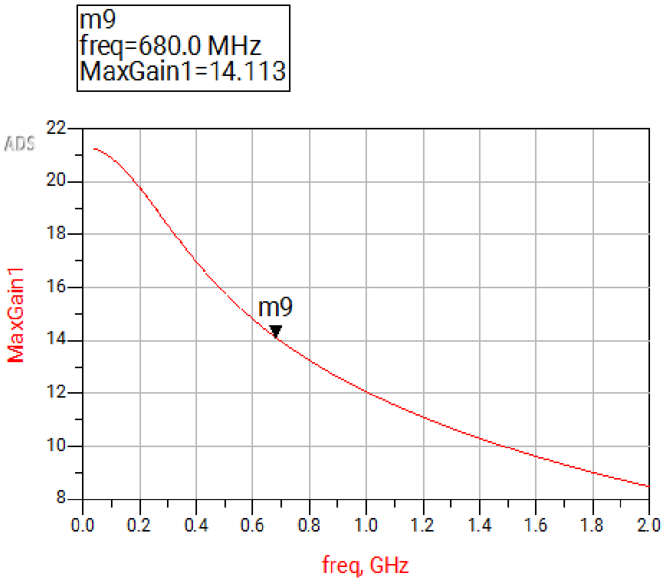

Figure 3: Maximum Available Gain Observed ≈14dB in the center frequency

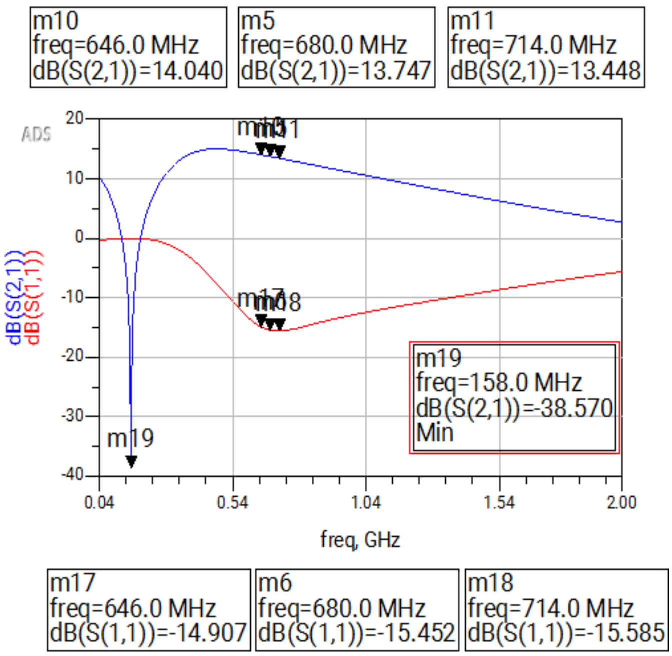

Figure 4: S21 Together with the S11 value

At the center frequency we get **13.75dB** gain which is highly above the required gain. At the lower limit of the bandwidth we get **14.04dB** and at the higher limit we get **13.45dB**. So, we have a small gain variation <0.5dB which satisfies the specification.

Input return loss is highly above 8dB in the band (approximately **15 dB**) (See Figure 4) The worst point in the band is 646MHz with **14.9dB**. In all frequencies the worst point is close to 158 MHz range. Output return loss is approximately **14dB** which is again above the 12dB limit. See the worse point is at upper limit of the band (Figure 5). At the lower limit of the bandwidth we get **14.97dB** and at the higher limit we get **13.82dB**.

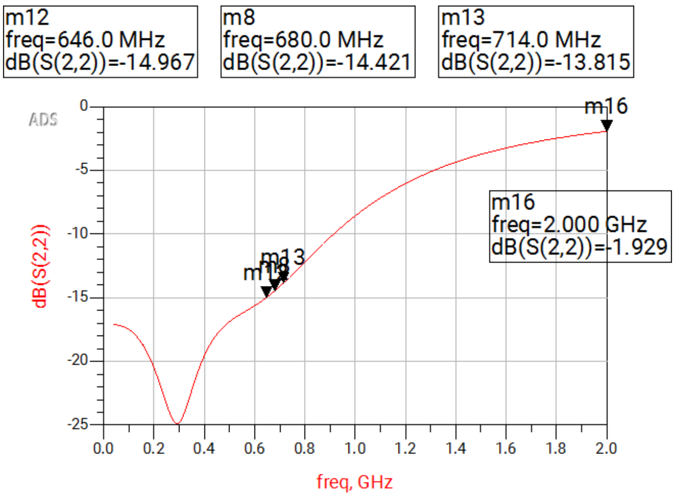

Figure 5: S22 Plot

For input output matching circuits Smith Chart helped. See Figure 6, my center frequency is close the the center which ensures the good matching.

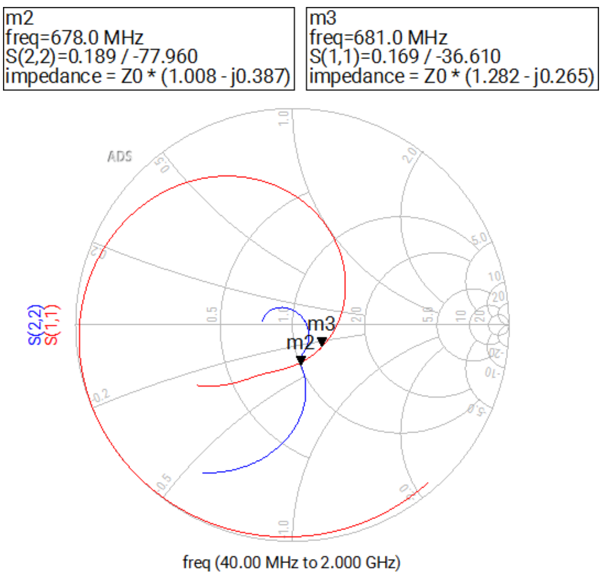

Figure 6: S22 and S11 on Smith Chart

The noise figure has to be little lower than our expected noise so that, the real noise figure will be lower enough, if NF is already higher than our excpected value we cannot reach a lower value whatever we are desiered to get. See below Figure gives NFmin about 0.8dB. Our NF has to be lower than **0.8244 dB.** See the worst point in the band is also below the range and at exactly on the center frequency we get **0.81dB.**

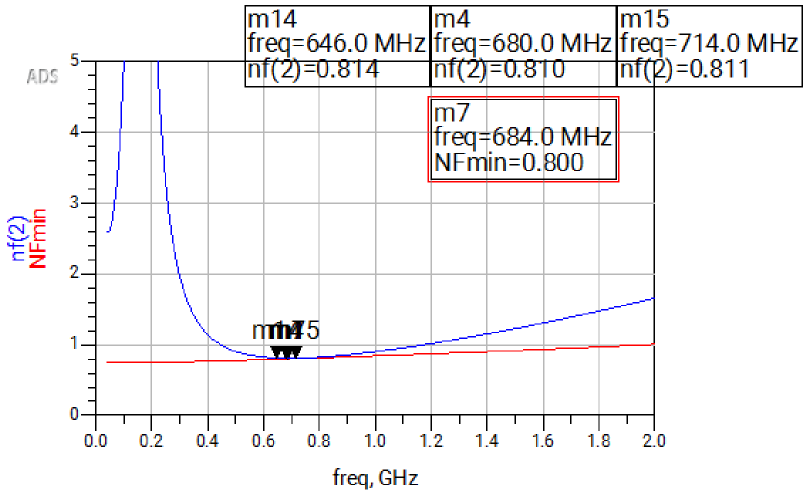

Figure 7: Noise Figure Plot in the Band

For determining Noise Figure and input output matching networks Figure 6 and 8 helped Figure 8 shows the circles at the center frequency

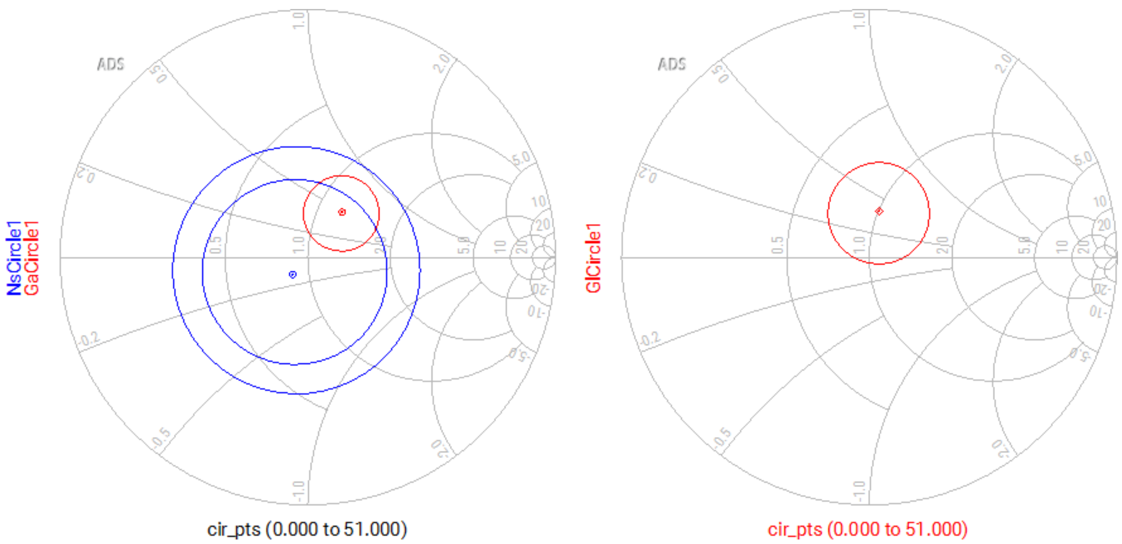

Figure 8: Noise, Ga and Gl Circles at 680MHz